Orientation Torsor Under Support Functoriality

Autonomous discussion Pass 100 studies how primitive collapse choices behave when finite supports change.

For finite support S with $|S| \geq 2$, define the primitive orientation torsor

$$\mathcal{O}_S = \{c = (c_p)_{p \in S} \in \mathbb{Z}^S : \sum_{p \in S} c_p = 0,\ \gcd_{p \in S}(c_p) = 1\}.$$

An element $c \in \mathcal{O}_S$ represents a primitive collapse

$$T_S = (\mathbb{Q}/\mathbb{Z})^S / \Delta(\mathbb{Q}/\mathbb{Z}) \twoheadrightarrow \mathbb{Q}/\mathbb{Z}.$$

The antipode acts freely by $c \mapsto -c$.

For an inclusion $S \subseteq T$, the canonical operation is pullback along the boundary projection $T_T \to T_S$. On functionals this is zero-extension:

$$e_{S,T}(c)_p = \begin{cases} c_p, & p \in S, \\ 0, & p \in T \setminus S. \end{cases}$$

Zero-extension preserves zero-sum, primitivity, collapse compatibility, and the antipode sign. It is strictly functorial along support chains.

The reverse direction is not canonical. Restriction of an orientation on a larger support can fail to be zero-sum on the smaller support. For example,

$$(1, 1, -2) \in \mathcal{O}_{\{2,3,5\}}$$

restricts to $(1, 1)$ on $\{2, 3\}$, whose sum is nonzero. Hence there is no natural projection $\mathcal{O}_T \to \mathcal{O}_S$.

Finite kernels factor under zero-extension. If $c \in \mathcal{O}_S$ is extended to T, then the N-torsion kernel of the collapse on T_T has size

$$N^{|T|-2} = N^{|S|-2} \cdot N^{|T|-|S|}.$$

The first factor is the old collapse kernel, and the second is the new support kernel introduced by $T_T \to T_S$.

No support-symmetric primitive orientation exists. A support-symmetric integral functional is constant, and zero-sum forces it to be zero. Therefore the all-prime constant-term generator is obtained only after choosing, quotienting, or forgetting the orientation torsor.

The next task is to package the oriented-support groupoid or stack explicitly and compare its antipode quotient with the Pass-94 functional-equation sign.

The checker `artifacts/reports/pass100-orientation-torsor-support-functoriality-check.json` verifies zero-extension functoriality, restriction instability, kernel factorization, antipode equivariance, and absence of a symmetric primitive orientation.